

ASSESSMENT DECLARATION COVER SHEET**STUDENT (REGISTRATION) NUMBER: 40804628****MODULE TITLE: Web Technologies****DATE OF SUBMISSION: 6-03-2026****Site Description**

The website is a fun game website called MP4orce. It is a turn-based game using HTML, CSS for designing and JavaScript for the game logic. The player has to play as a superhero team called MP4orce. The MP4orce is founded with 4 color-coded heroes. They have to deal with different types of enemies throughout different levels. They all have unique skills and player needs to use them wisely in order to defeat the enemies. The game also uses the XP system where player needs to collect XP to use skills or heal. Browser local storage will be used to save the progress the signed in players.

Research Summary

Clair Obscur Expedition 33

The idea of creating a turn-based game came from playing this game although it is not a JavaScript game.

FreeCodeCamp

This website is used for learning JavaScript deeply. Free Code Camp also gives the learner a certificate if they completed all 1059 steps.

YouTube

For watching tutorial and videos about how a developer or developers make a JavaScript game fun and attractive.

Channels: The Bald Coder, Goodgis

Coding Nepal

On this website, ten different browser-based games which are built using HTML, CSS and JavaScript can be learned. As it contains essential programming concepts without the need of external libraries, it is convenient to learn. It also provides live demos, video tutorials and source codes.

All of these reference site really helped to improve the problem-solving and logical thinking skills.

A collection of reference link can be shown [here](#).

List of Features

SIGN UP

If the player is an unregistered user, the website will ask for the user to sign up to the game. Local storage of browser will be used here to save the user data and progress.

SIGN IN

If the player is an already registered user, the website will ask for the user to fill the username and password in order to log on to the game.

ADJUST SETTINGS

Player can adjust settings like difficulty settings, effects like damage texts and screen shake, Language and music and background sound effects.

SAVE PROGRESS

This feature is added to save the game progress so the player does not have to start over when refresh the game or enter the game again.

VIEW THE BRIEF ABOUT HEROES

Player can also read about each hero and learn about their unique powers and skill in order to understand the game well.

GAME FEATURES

Turn-Based Combat

- Heroes and enemies take turns in an interleaved queue.
- Dead units are automatically skipped, keeping the pace tight.
- Attack animations slide units forward when they strike.
- Screen flash and particle bursts fire on every hit.

Skills & Abilities

- Each hero has 4 unique skills with different effects.
- Basic attacks are always free — advanced skills cost XP.
- Some skills guarantee a critical hit for massive bonus damage.
- Skills are locked behind XP cost to reward smart play.

XP System

- Every action earns +1 XP for the acting hero.
- Spend XP to unlock stronger skills mid-battle.
- Each hero tracks their own XP independently.
- XP resets at the start of each chapter, keeping each Figureht fresh.
- Skills cost 0 – 4 XP depending on power level.

Damage & Critical Hits

- All attacks roll a random variance — no two hits deal the same damage.
- 8% base critical hit chance for heroes on any attack.
- Enemies have a 15% critical chance.
- Critical hits deal $\times 1.5$ damage with a gold flash and larger number.
- Certain hero skills guarantee a critical when used.

Enemy AI

- Enemies target your lowest HP hero
- Enemy stats scale +12% per chapter to keep you challenged
- Enemies have their own critical chance

Chapter Progression

- Multiple chapters with escalating enemy counts and difficulty
- Chapter I starts with 1 enemy — Chapter IV throws 4 at once
- Final chapter is a solo boss Figureht.
- Clear all chapters to beat the game and see a scoreboard.

Visual Effects & UI

- Floating damage numbers pop off every hit.
- Heal numbers float up in green from the hero who recovers HP.
- XP gain floats up in purple after each action.
- Hit-shake animation rattles the target card on impact.
- HP bars use each hero's signature color.
- Active unit is highlighted with a gold glow and scale effect.
- Enemy targeting mode pulses red.

Navigation Diagram (Sitemap)

As soon as user enter the website, a menu page will welcome the user. The menu page will contain the navigation buttons to start playing the main game, adjust settings, sign in if the user is already sign up once and sign up if user is a completely new user. For the game, the game will have different levels where the heroes Figureht with different amount or different types of enemies. A scoreboard will be displayed as soon as the final chapter is beaten. User can also view the explanations about each hero i.e.; about their skills and attributes. User can also explore more about the website as the about us page and contact if desired.

Sitemap Diagram Image => [Appendix 1.1](#)

User Interface Wireframes => [Appendix](#)

Project Planning

This project is scheduled to run from January 29, 2026, through April 30, 2026. The project starts with Concept & Planning phase to define the project scope like choosing a suitable name for the website, feature list, and background research of similar sites. This leads directly into a Design stage where navigation diagrams, wireframes are created. UI style decisions are finally made to ensure an organized user experience. Technical construction is broken down into parallel tracks. These contain building a semantic multi-page HTML structure, establishing a robust CSS design system for responsive layouts, and programming the Core JavaScript Game Logic. This logic phase will be the most intensive part of the timeline as it covers the core mechanics of the games and match conditions over a three-week period.

An enhancement phase will be followed by the primary development to integrate Additional Features such as local storage for data persistence, custom animations, and sound effects. Testing, specifically focusing on cross-browser compatibility and UI adjustments will be done to ensure the game is well polished without any bugs. The project will be concluded with a code cleanup and official Deployment via GitHub Pages. Throughout this timeline, the schedule accounts for several holiday breaks in February, March, and April which are national holidays in Myanmar.

Gantt Chart can be viewed [here](#).

Appendix

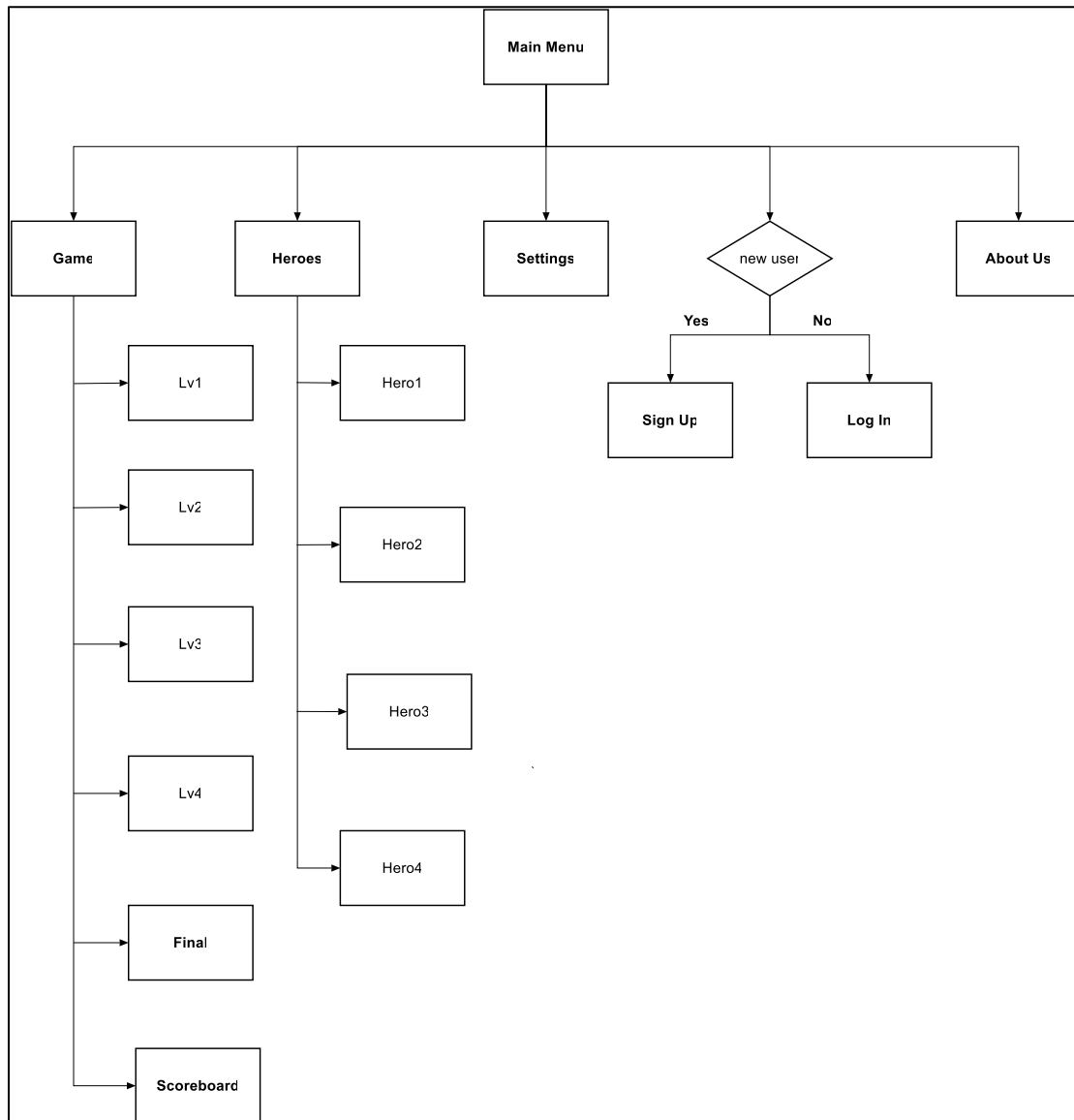


Figure 1.1: Sitemap for MP4orce

UI Wireframe for MP4orce

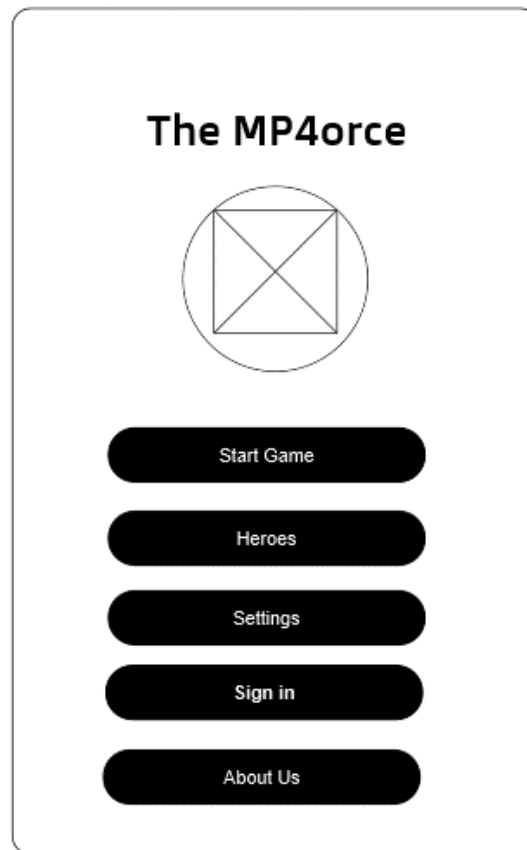
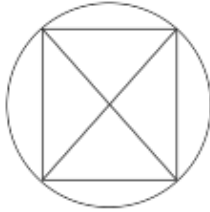


Figure 1.2: Main Menu Wireframe

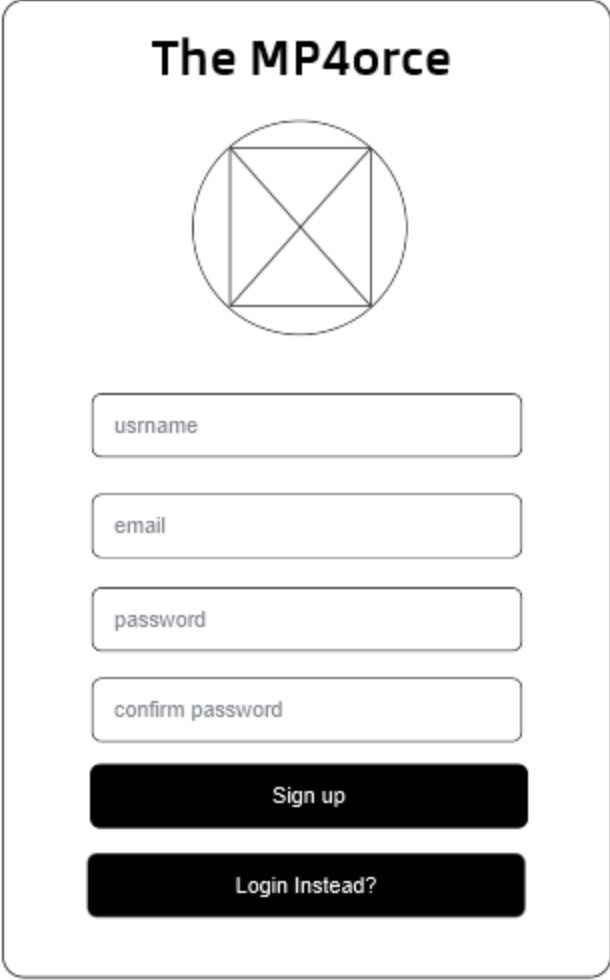
The MP4orce



Login

Don't have an account? Sign up

Figure 1.3: Login Wireframe



The MP4orce

A wireframe for a sign-up form titled "The MP4orce". The form is contained within a rounded rectangle. At the top is a logo consisting of a circle with a square inside, and the square has an 'X' formed by its diagonals. Below the logo are four text input fields stacked vertically, each with a placeholder label: "username", "email", "password", and "confirm password". At the bottom of the form are two dark buttons with white text: "Sign up" and "Login Instead?".

username

email

password

confirm password

Sign up

Login Instead?

Figure 1.4: Sign Up Wireframe

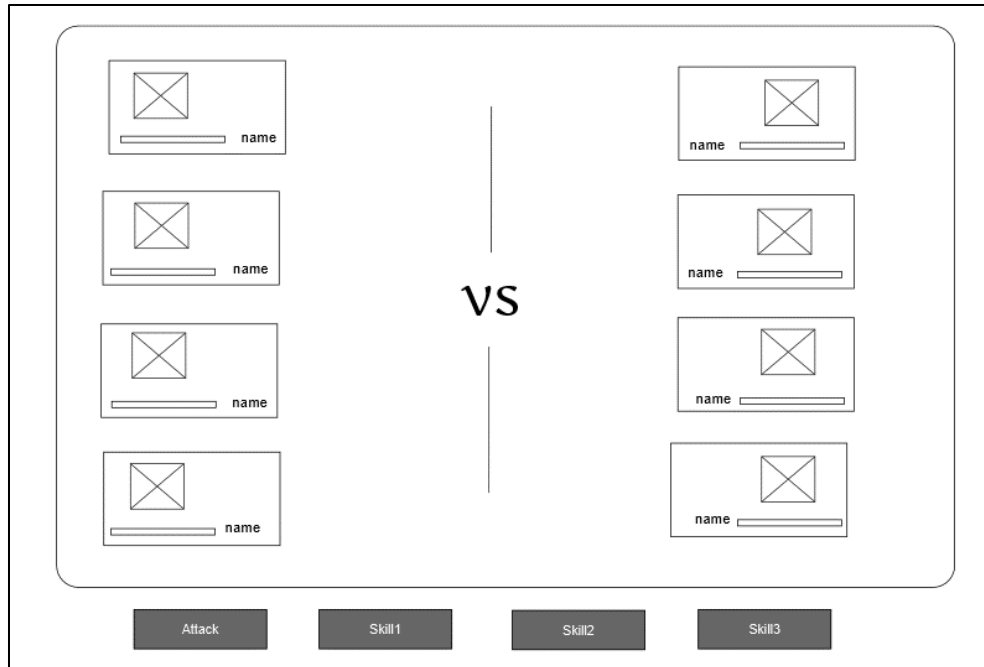


Figure 1.5: Main Game page Wireframe

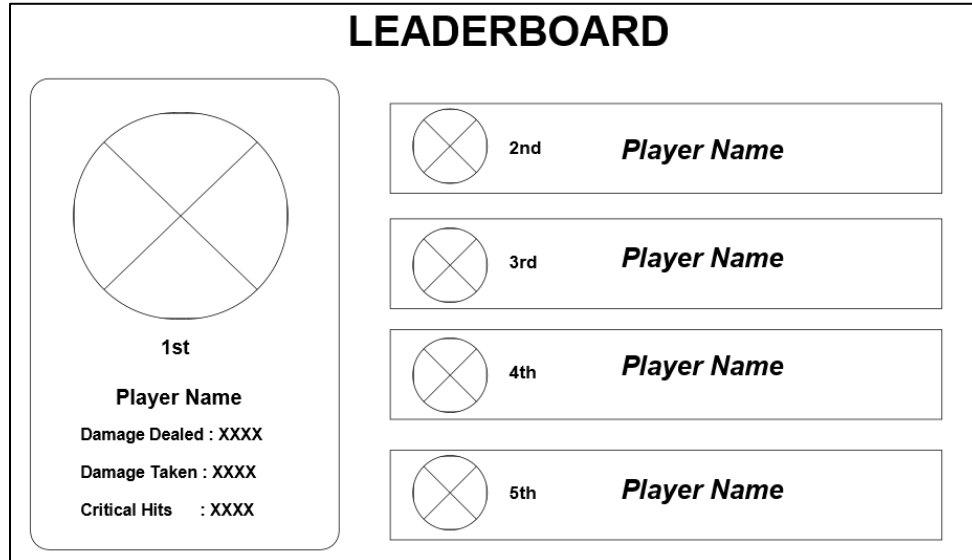


Figure 1.6: Leaderboard Wireframe

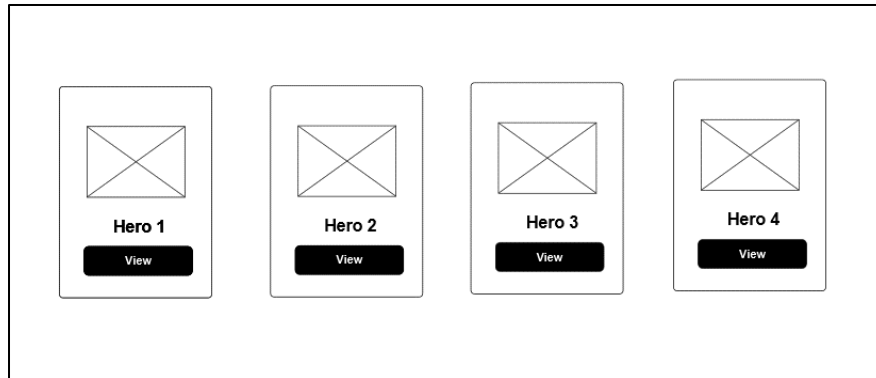


Figure 1.7: Heroes Page Wireframe

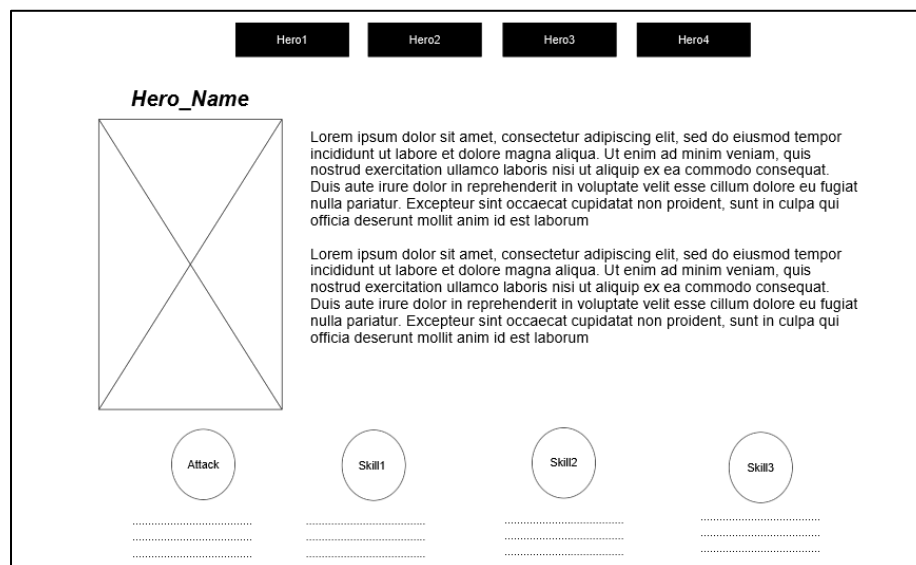


Figure 1.8: Hero Page Wireframe

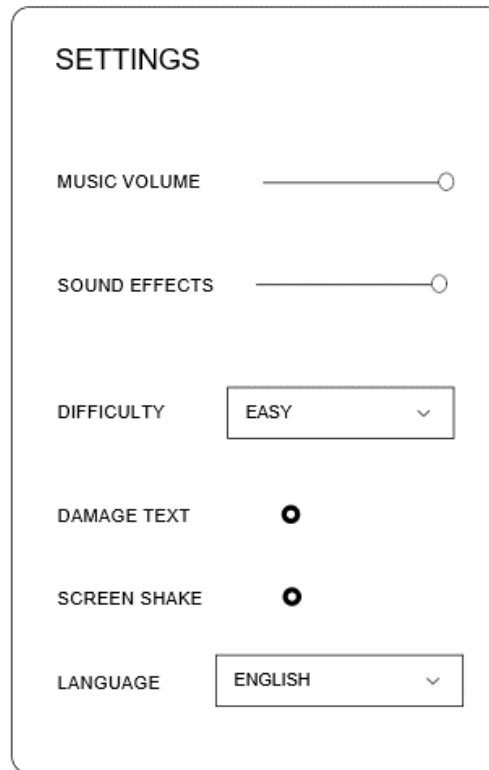


Figure 1.9: Settings Page Wireframe

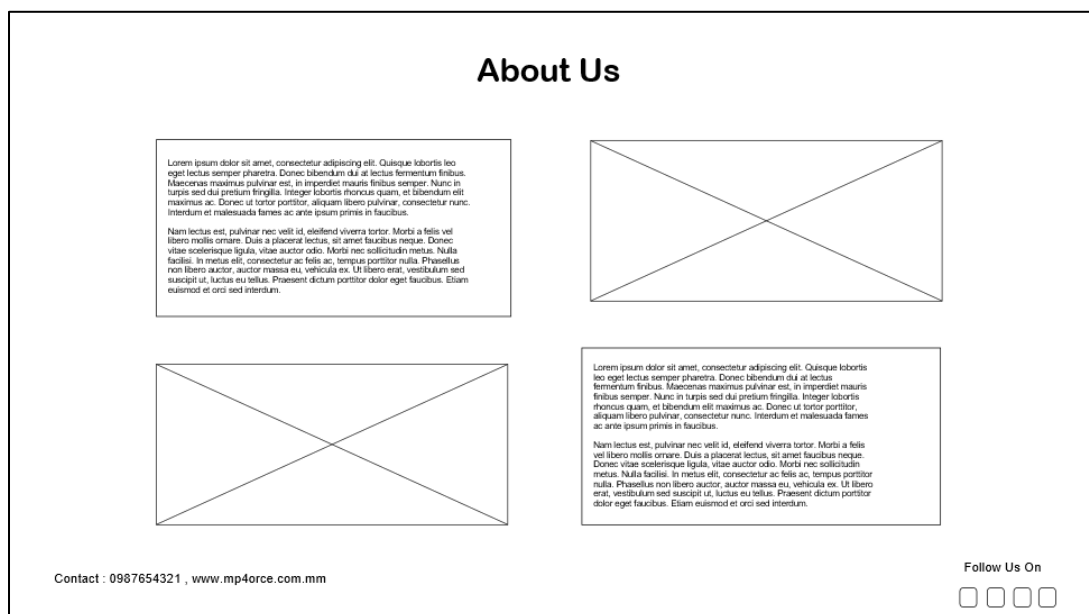


Figure 1.10: About Us Wireframe

The screenshot shows a Gantt chart in Microsoft Project. The timeline spans from 1/1/2018 to 1/1/2020. The chart displays various tasks with their durations and dependencies. Key tasks include 'Project Start', 'Project End', 'Project Kick-off', 'Project Planning', 'Project Execution', 'Project Monitoring', 'Project Reporting', 'Project Review', 'Project Closeout', and 'Project Handover'. The chart is color-coded by task type: blue for planning, green for execution, yellow for monitoring, and red for reporting. A vertical red line indicates the current date, which is approximately 1/1/2019.

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References

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